

Arbab Khan Lodhi

Mechatronics Engineering Student | Embedded Systems and Robotics

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National University of Sciences and Technology (NUST), College of EME, Pakistan



PROFILE

Third-semester Mechatronics Engineering student at NUST College of Electrical and Mechanical Engineering (EME), with hands-on experience in embedded systems, circuit projects, and CAD modeling: an ESP8266-based smart home automation system, an RLC circuit project, and a robot gripper model. Strong foundation in C++ programming and 3D CAD modeling, focused on embedded systems and automation.

EDUCATION

B.E. Mechatronics Engineering

2025 - 2029 (expected)

National University of Sciences and Technology (NUST), College of Electrical and Mechanical Engineering (EME)

- Currently in 3rd semester. Relevant coursework: Data Structures and Object-Oriented Programming (C++), Electronic Devices and Circuits, Engineering Dynamics, Materials and Manufacturing Processes, Linear Algebra.
- PEEF Centre of Excellence (COE) scholarship holder for the full bachelor's program.

PROJECTS

Smart Home Automation System

ESP8266 (NodeMCU) | Wi-Fi based appliance control

- Built an Internet of Things (IoT) system on the ESP8266 platform to switch and monitor home appliances remotely over Wi-Fi, with relay interfacing and sensor integration, as part of a team project.

RLC Circuit Project

Rectifier-RLC Filter | Circuit design and analysis

- Worked on an RLC circuit project based on a rectifier-RLC filter, covering circuit assembly and analysis as part of shared coursework/project work.

Robot Gripper Model

AutoCAD | Gripper mechanism modeling

- Modeled a robot gripper mechanism in AutoCAD, focusing on the gripper geometry and mechanical layout as part of shared CAD/project work.

TECHNICAL SKILLS

Programming: C++ (object-oriented programming, data structures), Arduino IDE

Embedded systems: ESP8266 (NodeMCU), ESP32, Arduino-based development, sensor and actuator interfacing, DC motor and relay control

CAD and design: SolidWorks part modeling and sketching; AutoCAD modeling, including a robot gripper mechanism

Tools: Git and GitHub, Visual Studio Code, web development fundamentals (React, HTML/CSS)

Languages: Urdu (native), English (fluent - academic and professional use)

CERTIFICATIONS

Career Essentials in Generative AI (Microsoft and LinkedIn); **Intro to Machine Learning** (Kaggle); **Open Science 101 and Open Science Essentials** (NASA); **Project Manager Job Simulation** (Siemens Mobility); **Cyber, Data Analytics, and Technology Job Simulations** (Deloitte Australia).

ACHIEVEMENTS

PEEF Centre of Excellence (COE) Scholarship

2025 - 2029

Punjab Educational Endowment Fund (PEEF)

- Awarded the PEEF COE scholarship for the Bachelor's Degree Program 2025-2029, in recognition of Intermediate Examination 2025 results.